Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional

Mid Report

Project Title : Farm Droid

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Farm Droid

An Autonomous Agricultural Robot

1. Introduction

Agriculture is facing significant challenges due to labor shortages and the need for precision monitoring of crop health. “Farm Droid” is a prototype for a compact, autonomous agricultural robot designed to address these issues through smart automation.

Unlike traditional heavy machinery which is expensive and inflexible, Farm Droid utilizes a lightweight, terrain-adaptive mobile platform. This allows it to navigate through crop rows efficiently to perform monitoring and manipulation tasks. The system integrates modern embedded technologies, using a Raspberry Pi 4 for computer vision and decision-making, and an Arduino UNO/ nano for real-time motor control and sensor data acquisition.

2. Objectives

The primary goal is to build a smart farming assistant capable of autonomous operation. The specific objectives are:

- **Environmental Monitoring:** To integrate sensors for measuring soil moisture, pH levels, and weather conditions (Rain/Temperature).
- **Robotic Manipulation:** To develop a servo-driven arm mechanism for different modular tasks like weeding.
- **Computer Vision:** To utilize a Raspberry Pi Camera and OpenCV to recognize crops, weeds, and obstacles in real-time.
- **Seamless Connectivity:** To establish reliable remote monitoring and control through telegram bot integration or a custom mobile app.
- **Autonomous Activity:** To allow the robot to perform tasks like plant seeding, watering, and soil fertilization without human intervention.
- **Adaptive Mobility:** To develop a robust locomotion system (Rover or Legged) capable of traversing uneven agricultural terrain.

3. Problem Statement

Conventional heavy farming machinery is often unsuited for small-scale precision agriculture or delicate crop fields. Large tractors cause soil compaction and lack the agility to perform individual plant inspection. There is a lack of affordable, modular robotic solutions that can perform both monitoring and physical manipulation tasks in such environments. Farm Droid aims to bridge this gap by providing a lightweight, intelligent, and versatile solution.

4. Scope

The project scope encompasses the mechanical assembly, circuit integration, and software development of the robot.

- **Hardware:** Custom chassis design, Motor/Servo drive system, Power distribution system.
- **Software:** Python-based vision processing, C++ based motor control firmware.
- **Limitation:** The prototype will focus on small-scale demonstration (e.g., a garden or test bed) rather than full industrial field deployment.

5. Methodology

5.1. System Architecture

We are adopting a “Distributed Computing” architecture to handle the computational load efficiently:

- **The Brain (Raspberry Pi 4):** Handles high-level logic, image processing (OpenCV), and path planning. It communicates via Serial/I2C/USB.
- **The Controller (Arduino/ESP32):** dedicated to PWM generation for the motor drivers and reading analog sensors.

5.2. Technology Stack

- **Language**: Python (Vision/Logic), C++ (Arduino Firmware)
- **Vision**: OpenCV, TensorFlow Lite (Optional for object detection)
- **Hardware Interface**: I2C (Inter-Integrated Circuit), USB.
- **Power Management**: A 65W/22W 30000mAh with Buck Converters for system stability.

6. Hardware Components

The core components for the Farm Droid include:

- **Actuators:** DC Gear Motors or MG996R Servos (Mobility), SG90 (Gripper/Arm), GA25 (Metal Gear motor).
- **Sensors:** Ultrasonic (HC-SR04), Soil Moisture, Rain Sensor, DHT11, IR (Infrared), LDR (Light Dependent Resistor), DFPlayer Mini, RFID Reader(Radio Frequency Identification), Relay, PH Electrode, Gas sensor(MQ135).
- **Controllers:** Raspberry Pi 4 Model B (4GB), Arduino Uno / Nano.
- **Drivers:** L298N/Motor Driver (for Wheels).
- **Vision:** Official Raspberry Pi Camera Module using Pi camera 5MP.
- **Power Supplier:** LM2596 Buck Converter, 65W/22W 30000mAh Power Bank.

7. Budget Estimation

The following is a detailed breakdown of the estimated costs for the hardware components required for the “Farm Droid” project based on real market prices in Bangladesh. The total estimated cost for the project is approximately BDT 31488. This budget includes all necessary components for building the prototype, such as controllers, sensors, actuators, power supplies, and miscellaneous materials.

Category	Component Name	Unit (BDT)	Qty	Total (BDT)
Controllers	Raspberry Pi 4 Model B (4GB)	14,100	1	14,100
	Arduino Uno R3	988	2	1,976
Motors & Drivers	GA25 Metal Gear Motor	450	2	900
	DSServo DS3218 20KG Digital Metal Servo with Horn 270 Degree	2350	2	4700
	Servo Motor MG996R	500	1	500
	Servo Motor SG90 (Micro)	180	2	360
Sensors	HC-SR04 Ultrasonic Sensor	99	1	99
	FC-51 IR Obstacle Sensor	45	3	135
	HC-SR501 PIR Motion Sensor	93	1	93
	OV7670 Camera Module	400	1	400
	DHT11	95	1	95
	LDR(Light Dependent Resistor)	5	1	5
	DFplayer	350	1	350
	PIR Sensor	93	1	93
	RFID	190	1	190
	Relay	80	4	320
	PH Sensor	2584	1	2584
	Moisturizer Sensor	70	1	70
	Rain Sensor	78	1	78
	MQ135	135	1	135
Power & Audio	Breadboard Power Supply	76	1	76
	3 Watt 8 Ohm Mini Speaker	250	1	250
	Electret Microphone	15	2	30
Mechanical & Misc	PVC Pipe	300	5	1,500
	Pipe (Small)	30	5	150
	Half-Size Breadboard	75	2	150
	Jumper Wires (Male-Male)	100	4	400
	USB Cable (Arduino)	150	2	300
	Screws / Hardware	100	10	1,000
	Copper Wire (26 SWG)	110	2	220
Tools	Digital Scale (10kg)	259	1	259
	Hex Saw	15	2	30
	Sand Paper	30	3	90
			Grand Total Estimated Cost	**31488**

Table 1: Detailed Budget Estimation

8. Work Plan (Timeline)

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Analysis & Planning	✓					
Mechanical Assembly (Chassis & Drive)	✓	✓				
Circuit Integration & Power Distribution		✓	✓			
Sensor Installation & Wiring		✓	✓			
Motor Control & Calibration			✓	✓		
Arm Control & Actuation Testing				✓	✓	
Software Integration				✓	✓	✓
System Testing & Debugging					✓	✓
Final Review & Documentation						✓

Table 2: Gantt Chart for Farm Droid Development

9. Visual Models

9.1. Circuit Diagram

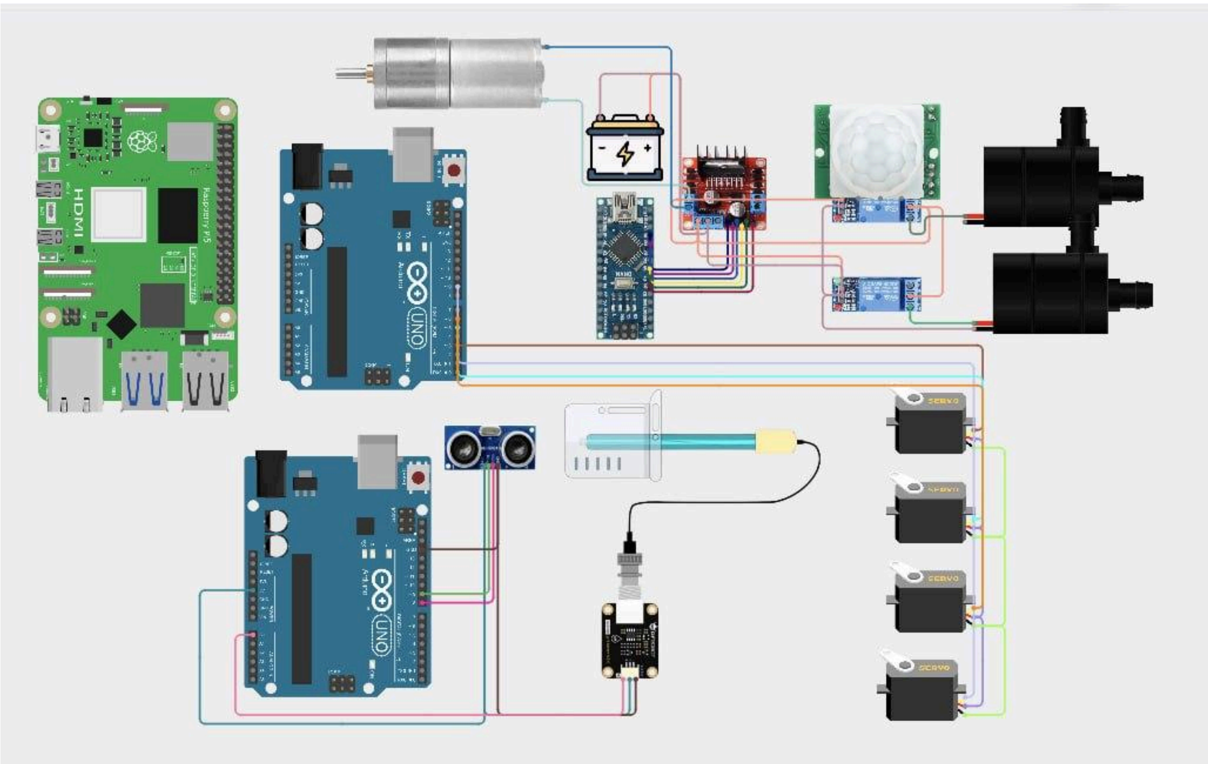

Figure 1: Circuit Diagram of Farm Droid System

9.2. Data Flow Diagram

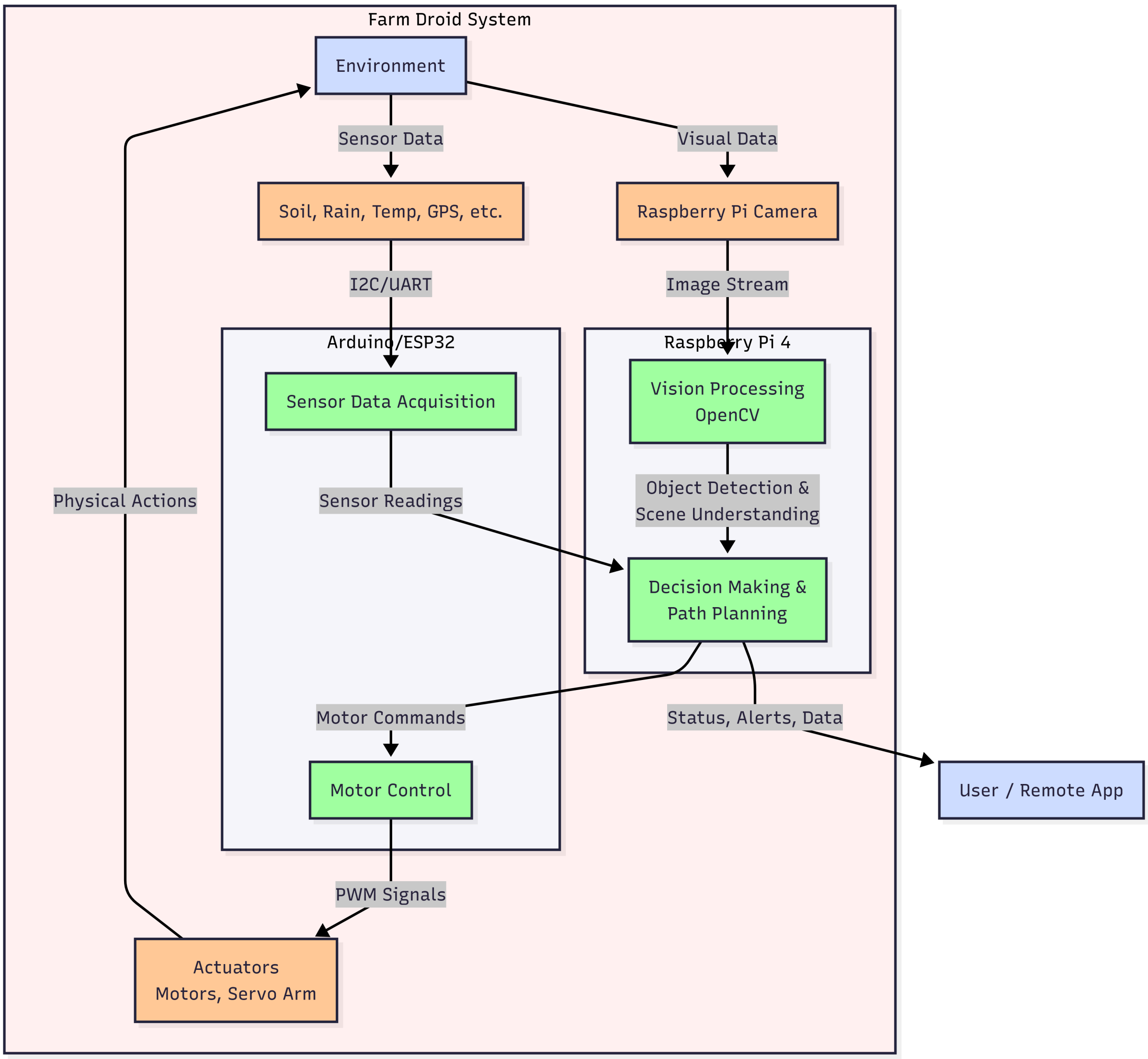

Figure 2: Data Flow Diagram of Farm Droid System

10. Future Plans

1. **Telegram Bot:** Implementing a Telegram bot for real-time alerts and control commands.
2. **Mobile App:** Developing an app to monitor the bot's status remotely.
3. **FastAPI:** Building a RESTful API for backend services.
4. **Computer Vision:** Implementing computer vision algorithms for object detection and recognition.

11. Conclusion

Farm Droid represents a step towards modernizing agriculture through robotics. By combining autonomous mobility with computer vision, this project aims to demonstrate that farming automation can be agile, intelligent, and accessible. The successful completion of this project will provide a functional prototype capable of navigating autonomous paths and performing basic agricultural tasks.